

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

21:25

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

Code:

```
import pandas as pd
```

```
import numpy as np
```

```
# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

data['FamilySize'] = data['SibSp'] + data['Parch']


# 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)


data['Alone'] = np.where(data['FamilySize'] == 0, 1, 0)

# 2. Convert 'Sex' to numeric (male: 0, female: 1)

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})


# 3. One-hot encode the 'Embarked' column

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)


# 4. Get the mean age of passengers

mean_age = data['Age'].mean()

print(mean_age)

# 5. Get the median fare of passengers

median_fare = data['Fare'].median()

print(median_fare)


# 6. Get the number of passengers by class

passengers_by_class = data['Pclass'].value_counts()

print(passengers_by_class)
```

7. Get the number of passengers by gender

```
passengers_by_gender = data['Sex'].value_counts().sort_index()  
print(passengers_by_gender)
```

8. Get the number of passengers by survival status

```
passengers_by_survival = data['Survived'].value_counts().sort_index()  
print(passengers_by_survival)
```

9. Calculate the survival rate

```
survival_rate = data['Survived'].mean()  
print(survival_rate)
```

10. Calculate the survival rate by gender

```
survival_by_gender = data.groupby('Sex')['Survived'].mean()  
print(survival_by_gender)
```

Average time
0.412 s
412.00 ms



Maximum time
0.412 s
412.00 ms



✓ 1 out of 1 shown test case(s) passed

✓ Test case 1

412 ms

Debug



Expected output

Actual output

29.69911764705882

29.69911764705882

14.4542

14.4542

3...491

3...491

1...216

1...216

2...184

2...184

Name: Pclass, dtype: int64

Name: Pclass, dtype: int64

0...577

0...577

1...314

1...314

Name: Sex, dtype: int64

Name: Sex, dtype: int64

0...549

0...549

1...342

1...342

Name: Survived, dtype: int64

Name: Survived, dtype: int64

0.3838383838383838

0.3838383838383838

Sex

Sex

0...0.188908

0...0.188908

1...0.742038

1...0.742038

Name: Survived, dtype: float64

Name: Survived, dtype: float64